Online JEPA Composite Stability
A Two-Time-Scale Extension of Borkar (2008)
for Recursive Controlled Systems with Learned Substrates

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Abstract

Scaffold placeholder. Paper 0 of the Recursive Controlled Systems series proves a per-level stability budget $\lambda_k = \gamma_k - L_\theta \rho_k - L_d \eta_k - \beta \bar{\tau}_k - \frac{\ln \nu}{\tau_a}{}_k$ under hand-set Lyapunov functions and hand-set constants. This paper generalises the budget to the setting in which the substrate V_k itself is *learned online* by a Joint-Embedding Predictive Architecture (JEPA), yielding two new cost terms: an *online encoder cost* $L_{o\theta}(\Delta, \mu)$ derived by extending Borkar's Ch. 6 two-time-scale stochastic approximation to discrete deployment events, and a *head cost* $L_H \, \kappa_k$ that bounds substrate sensitivity to head-specific payload variations. The new bound reduces to Paper 0's Theorem VI exactly when $\Delta \to \infty$ (frozen substrate) and $L_H \leq 1$ (non-amplifying head); otherwise it is strictly tighter and gates a CI-enforced production rollout via the empirical-constants protocol of Section 6. The final abstract will be written in T5 once the constants table from Q3 substrate measurements is in hand. See `PROTOCOL.md` for the locked theorem statement and proof skeleton.

Contents

1 Introduction

2 Related Work

3 RCS Recap

4 Theorem Statement

5 Proof

6 Empirical-Constants Estimation Protocol

7 Discussion

8 Conclusion

References

[1] Christopher M. Bishop. *Pattern Recognition and Machine Learning*. Information Science and Statistics. Springer, 2006. ISBN 978-0-387-31073-2. Ch. 10 (variational inference) is the standard reference for the KL-bound lemma supporting Paper 5 (H8d).

[2] Vivek S. Borkar. *Stochastic Approximation: A Dynamical Systems Viewpoint*. Cambridge University Press / Hindustan Book Agency, 2008. ISBN 978-0-521-51592-4. Ch. 6 is the foundational two-time-scale stochastic approximation reference; extended by Paper 5 T1 (BRO-989) to discrete deployment events.

[3] John N. Tsitsiklis and Benjamin Van Roy. An analysis of temporal-difference learning with function approximation. *IEEE Transactions on Automatic Control*, 42(5):674–690, 1997. doi: 10.1109/9.580874.